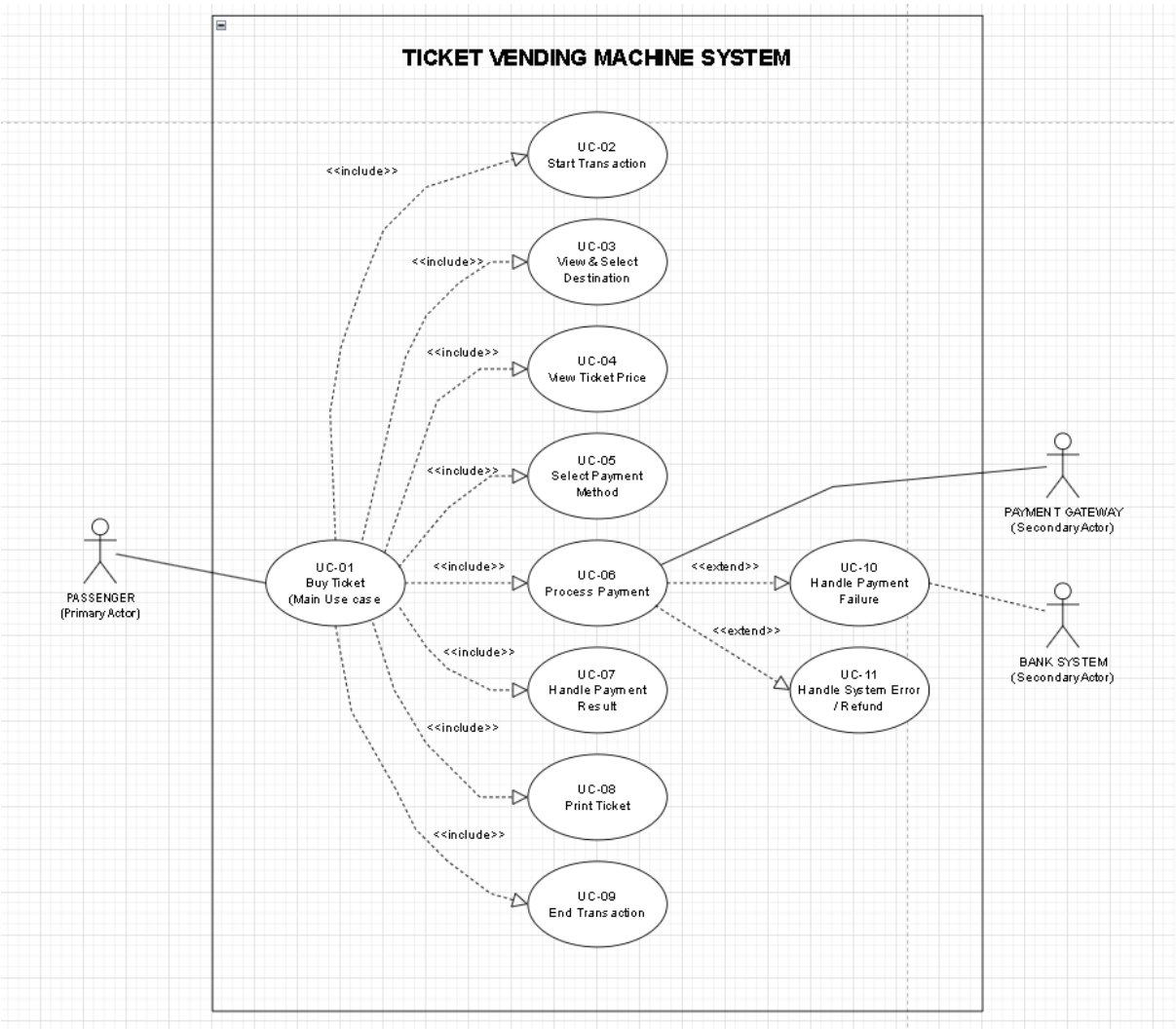
USE CASE SPECIFICATION: TICKET VENDING MACHINE SYSTEM

I. MAIN USE CASE: PURCHASE TRAVEL TICKET



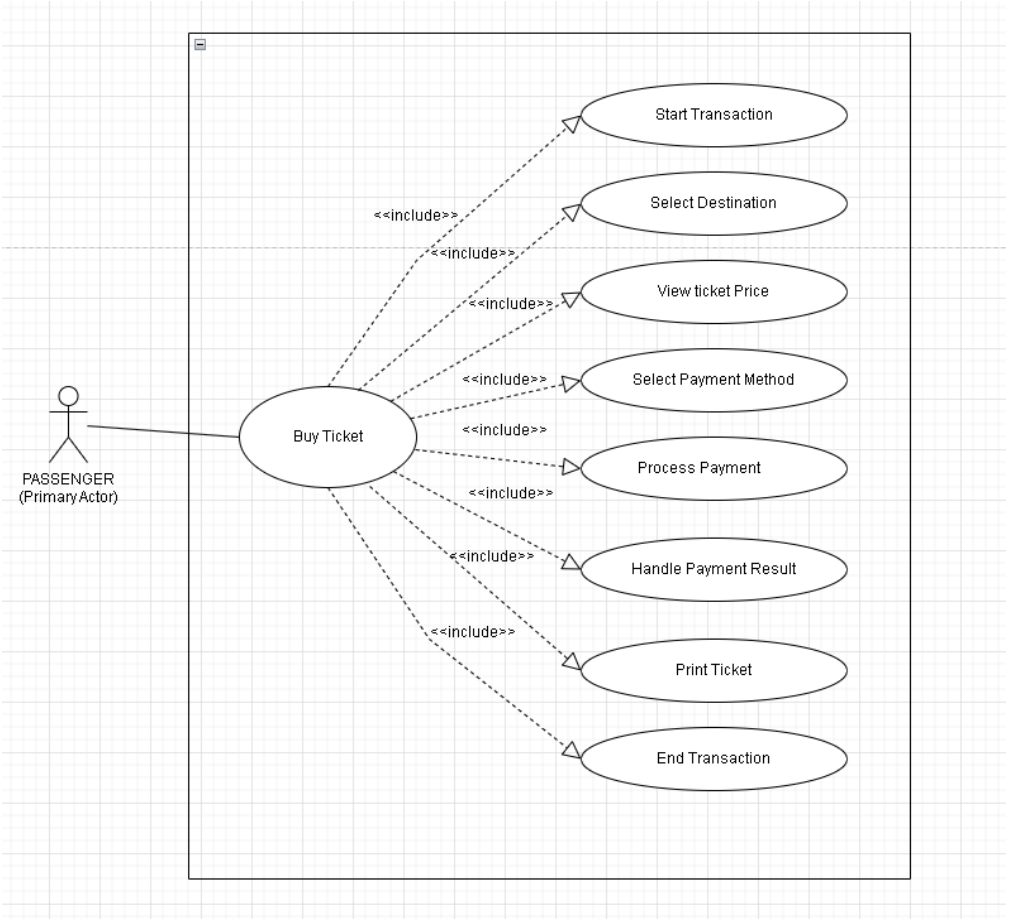
This is the high-level process representing the passenger's goal to obtain a ticket.

Item	Description
Use Case Name	**Purchase Travel Ticket (General)**
Primary Actor	Passenger

Secondary Actors	Payment Gateway, Bank System
Brief Description	The passenger uses the Ticket Vending Machine (TVM) to select a destination, pay the required fare, and receive a printed ticket.
Pre-conditions	The TVM is in "Ready" mode with sufficient paper and network connection.
Post-conditions	The passenger receives a valid ticket; the transaction is logged; the machine resets to Idle.
Main Success Scenario	1. Passenger initiates the process (UC-01, UC-02). 2. Passenger selects destination and views price (UC-03, UC-04). 3. Passenger chooses payment method and completes payment (UC-05, UC-06). 4. System handles result and prints ticket (UC-07, UC-08). 5. System ends the session (UC-09).

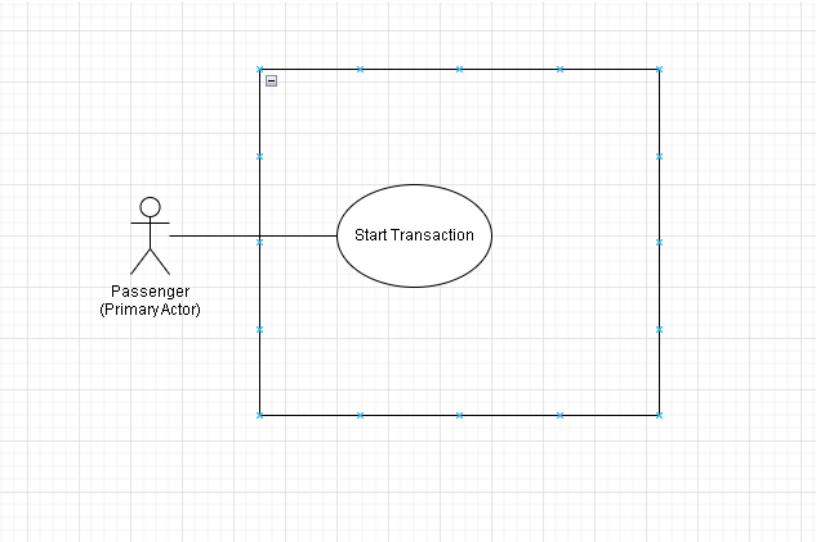
II. DETAILED DECOMPOSITION (11 USE CASES)

UC-01: Buy Ticket (Initiation)



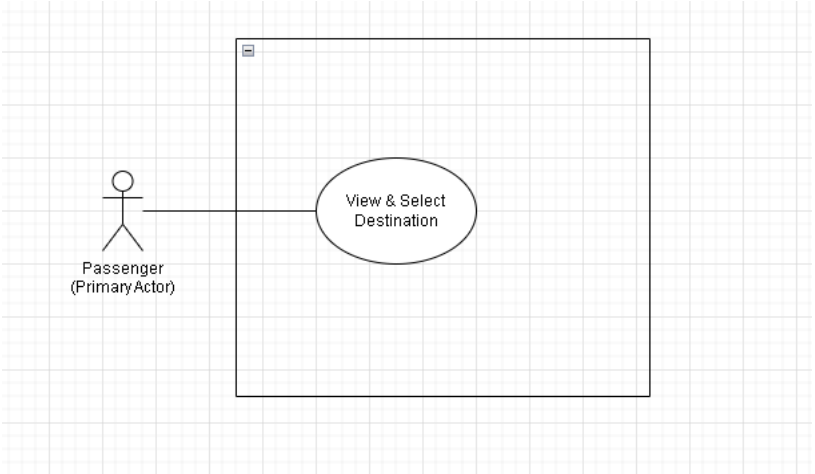
- **Description:** The entry point of the purchasing process where the user interacts with the "Buy Ticket" function on the main screen.

UC-02: Start Transaction



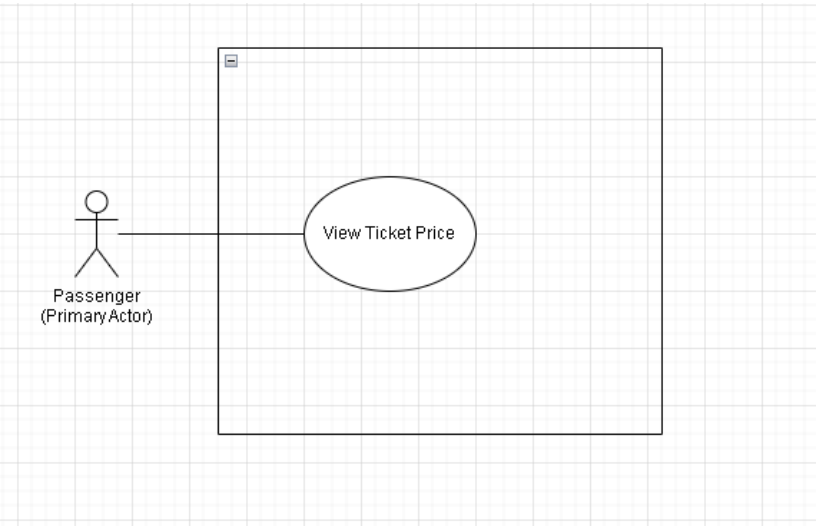
- **Description:** The system initializes the hardware and software environment for a new session, clearing any cached data from previous users.

UC-03: View & Select Destination



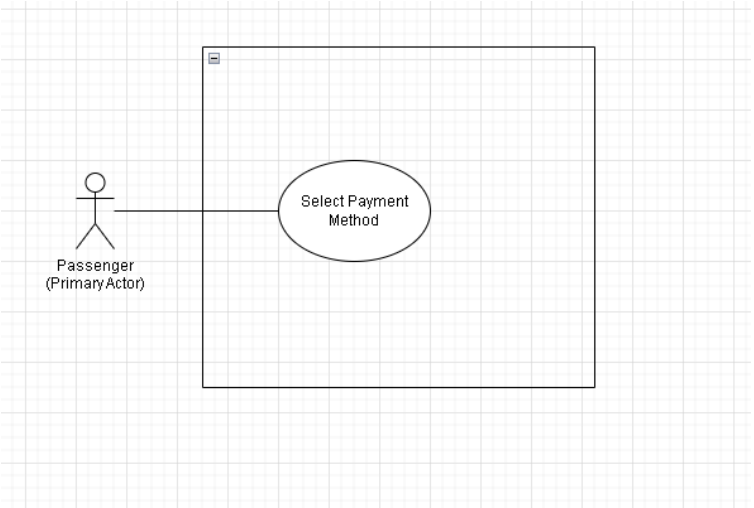
- **Description:** Displays a list of stations or a digital map. The passenger selects the specific destination for their trip.

UC-04: View Ticket Price



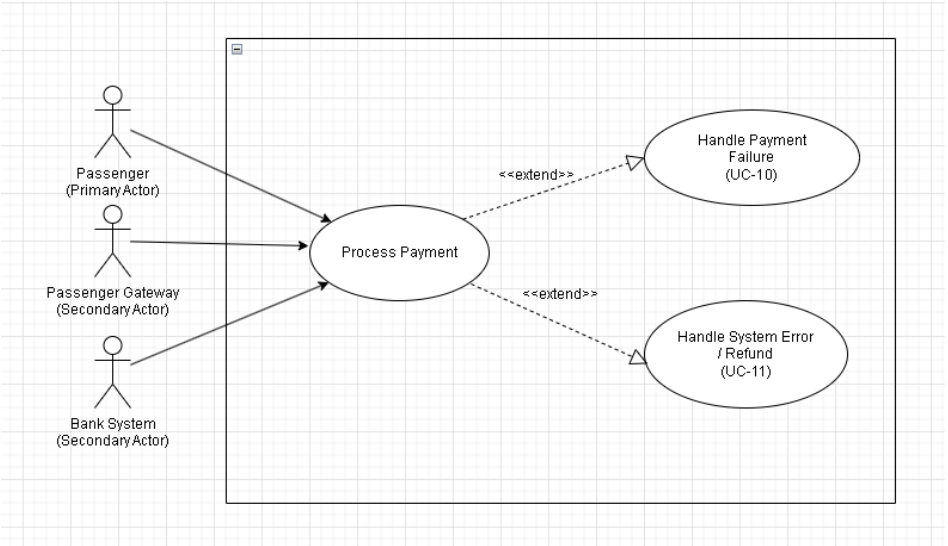
- **Description:** The system calculates the fare based on the current location and the selected destination, displaying the total amount due.

UC-05: Select Payment Method



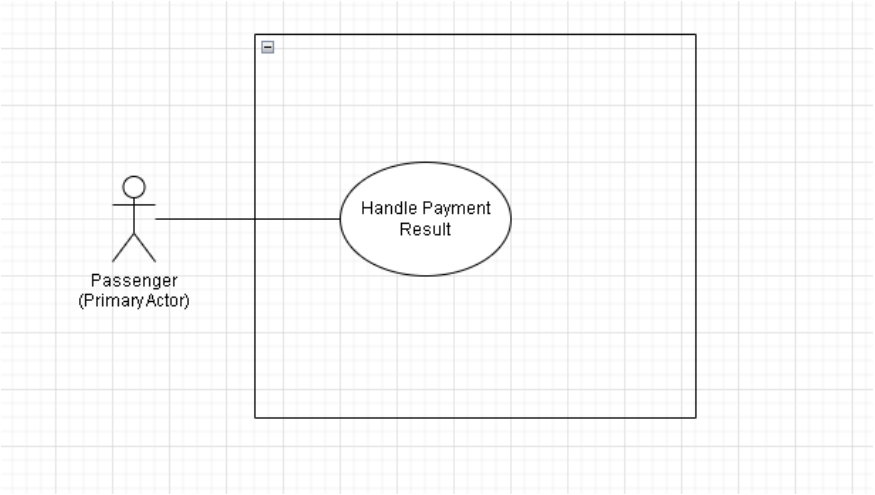
- **Description:** The passenger chooses between available payment options: Cash, Credit/Debit Card, or Mobile QR.

UC-06: Process Payment



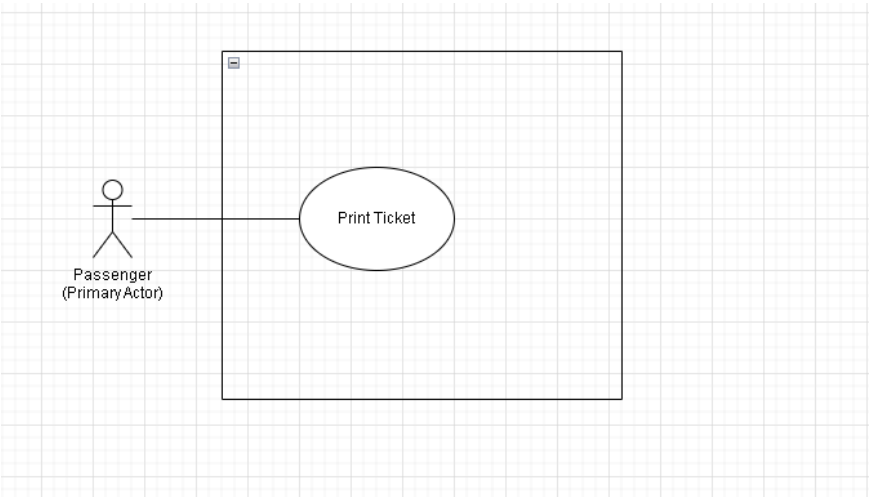
- **Description:** The system communicates with the **Payment Gateway** and **Bank System** to authorize and execute the fund transfer.
- **Extensions:** UC-10 (Failure), UC-11 (System Error).

UC-07: Handle Payment Result



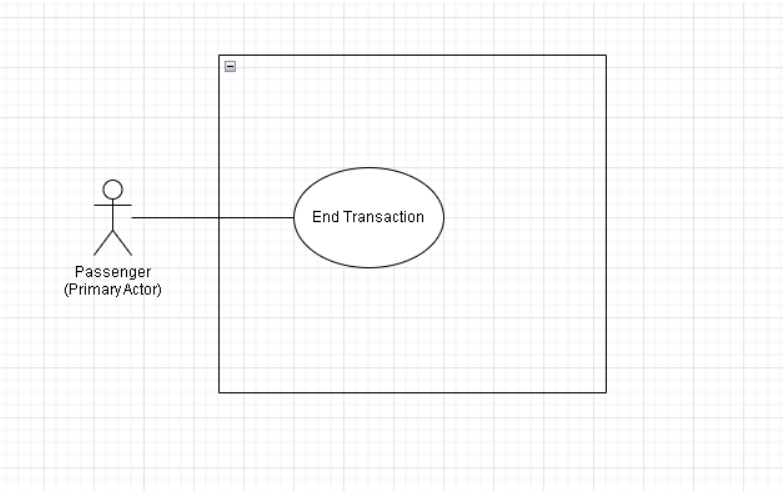
- **Description:** The system receives feedback from the bank. If successful, it moves to the printing stage; otherwise, it triggers error handling.

UC-08: Print Ticket



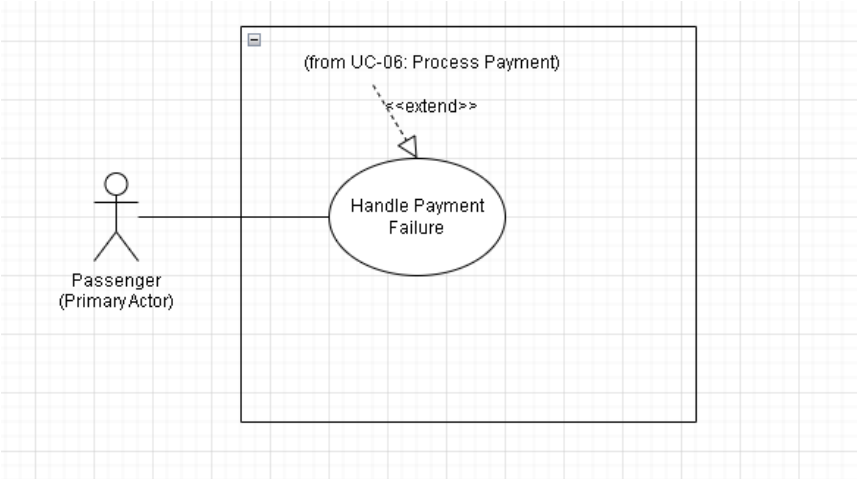
- **Description:** The internal thermal printer generates the physical ticket and delivers it to the collection tray.

UC-09: End Transaction



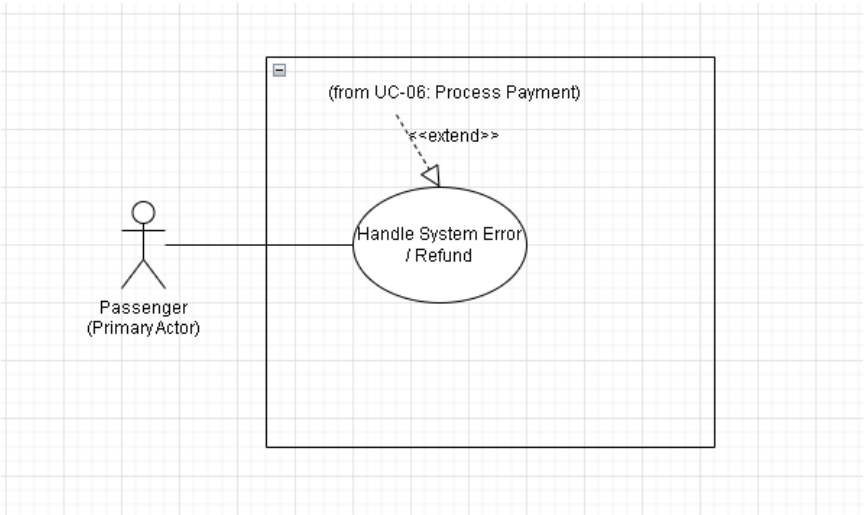
- **Description:** The system clears the current session, displays a "Thank You" message, and returns to the "Ready" state for the next user.

UC-10: Handle Payment Failure (Extension)



- **Trigger:** UC-06 returns a "Declined" or "Insufficient Funds" status.
- **Flow:** The system notifies the user and allows them to change the payment method or cancel.

UC-11: Handle System Error / Refund (Extension)



- **Trigger:** A hardware failure (e.g., printer jam) or network timeout occurs during UC-06.
- **Flow:** The system automatically reverses the charge (Void/Refund) and logs a maintenance report.